AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

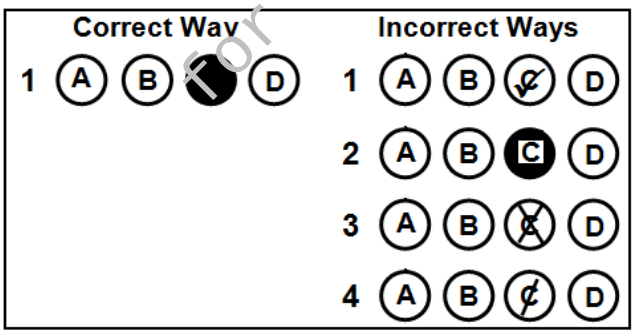
Business Statistics Paper I

Time: 55 minutes Marks: 30

AKU-EB Annual Examinations 2025 For Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meanings.

Measures of Central Tendency and Quartiles

$$\overline{X} = \frac{\sum wx}{\sum w} \qquad \overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{\sum f} \qquad \mathrm{Q}_i = l + \frac{h}{f}\left(\frac{n \times i}{4} - c\right)$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Measures of Dispersion

$$\text{variance} = \frac{\sum \left(x - \overline{X}\right)^2}{n} \qquad \text{SD} = \sqrt{\frac{\sum \left(x - \overline{X}\right)^2}{n}}$$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{SD} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{variance} = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2 \qquad SD = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

$$\text{variance} = \frac{\sum f\left(x - \bar{x}\right)^2}{\sum f} \qquad SD = \sqrt{\frac{\sum f\left(x - \bar{x}\right)^2}{\sum f}}$$

$$\mathrm{R} = x_{\max} - x_{\min} \qquad \text{Coefficient of Variance} = \frac{SD}{\overline{X}} \times 100$$

Index Number

$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100 \qquad P_{0n} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times 100 \qquad P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}} \times 100$$

Counting Techniques and Probability

$${}^{n}P_r = \frac{n!}{(n-r)!} \qquad {}^{n}C_r = \frac{n!}{(n-r)!r!}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(A \cap B) = P(A) \times P(B) \qquad P(A \cap B) = P(B) \times P(A \mid B)$$

Scatter Diagram and Correlation

$$r = \frac{n \sum XY - \left(\sum X\right)\left(\sum Y\right)}{\sqrt{\left[n \sum X^2 - \left(\sum X\right)^2\right]\left[n \sum Y^2 - \left(\sum Y\right)^2\right]}}$$

AKU-EB Annual Examinations 2025 for Teaching & Learning only

1. All of the following are the sources of secondary data EXCEPT

   A. websites.
   B. journal articles.
   C. personal interview.
   D. government record.

2. Which of the following is a continuous variable?

   A. The maximum number of people that can be seated in a hall
   B. The average age of students in a class of a school
   C. The total number of dots on all the faces of a die
   D. The number of siblings of Akram

3. One of the examples of discrete variable is

   A. weight of the students.
   B. gender of the students.
   C. height of the students.
   D. number of students.

4. The given frequency distribution represents the information about the number of pets owned by students in a class.

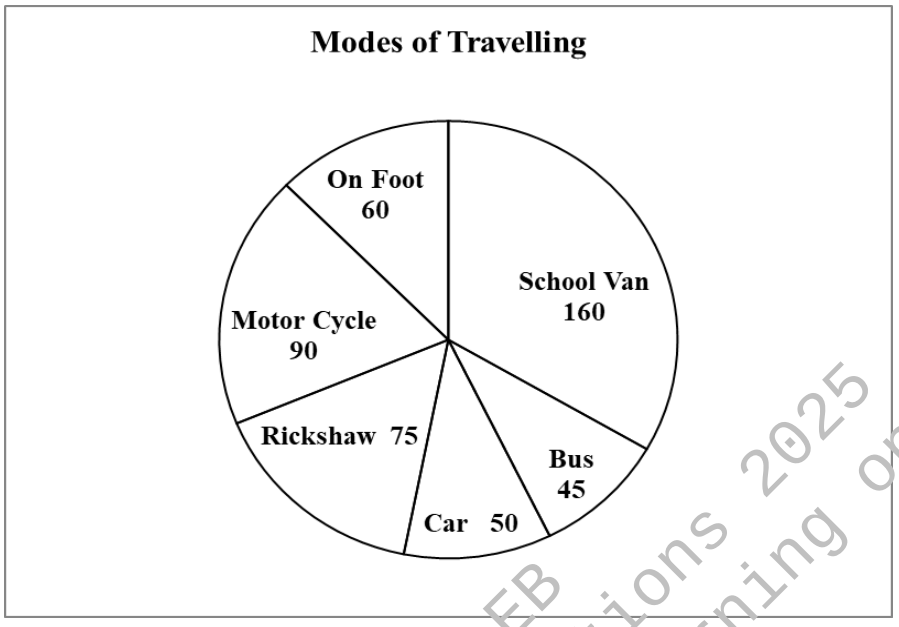
Number of Pets	Number of Students
0 – 1	8
2 – 3	7
4 – 5	9
5 – 7	5
8 – 9	1

   The class mark of the class interval with the least frequency is

   A. 0
   B. 7.5
   C. 8.5
   D. 11

AKU-EB Annual Examinations 2025 For Teaching & Learning only

5. For a school, the given pie chart represents the number of students and their modes of travelling to their school.



AKU-EB Annual Examinations 2025 for Teaching & Learning only

The central angle of the sector, representing the number of students which DO NOT travel by school van is

A. 240°
B. 160°
C. 120°
D. 33°

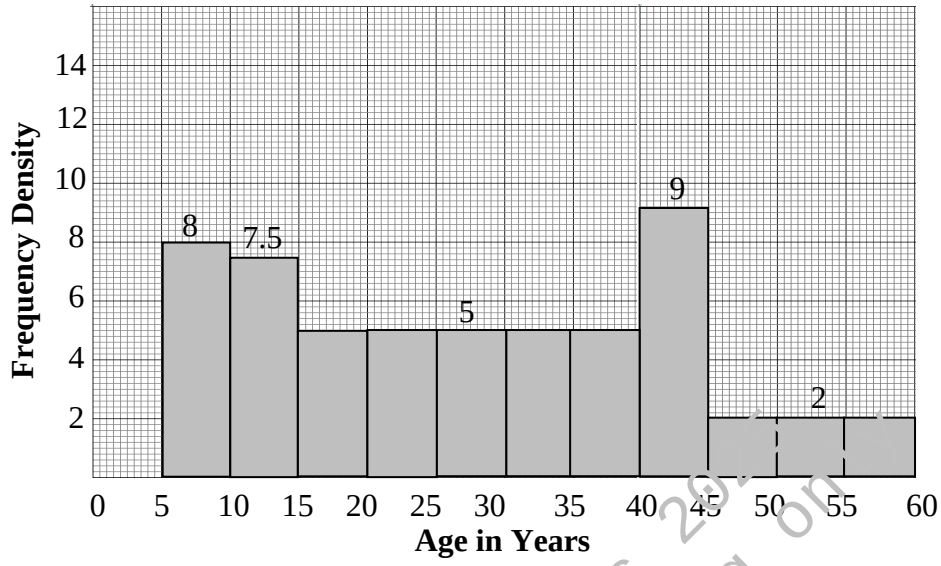
6. The given frequency table shows the LDL cholesterol of 30 persons aged between 25 to 30 years.

LDL Range mg/ dL	Number of Persons
60 – 69	2
70 – 79	6
80 – 89	12
90 – 99	7
100 – 109	3

The percentage of persons whose LDL cholesterol ranges from 80 to 89 mg/ dL is

A. 10%.
B. 12%.
C. 40%.
D. 60%.

7. The given histogram shows the age-wise distribution of residents of a colony.



The number of residents that are older than 45 years of age will be

A. 2
B. 9
C. 30
D. 45

8. Which of the following options can be CORRECTLY represented by Pie chart?

I. Time
II. Gender
III. Qualification

A. I only
B. II only
C. I and III
D. II and III

9. If the mean of a grouped data is 35 and $\sum fx = 1505$, then $\sum f$ is equal to

A. 43
B. 50
C. 1,470
D. 52,675

10. If the mean ($\overline{X}$) of a variable X is 30, then the mean of variable $\frac{2X}{3} + 5$ will be

A. 20
B. 25
C. 30
D. 50

AKU-EB Annual Examinations 20... for Teaching & Learning on...

11. The arithmetic mean of a data set consisting of 10 values is *x*. If another data set consists of 15 values and has arithmetic mean as *y*, then the combined mean of the two data sets will be

A. $\frac{15x+10y}{2}$

B. $\frac{10x+15y}{2}$

C. $\frac{10x+15y}{25}$

D. $\frac{15x+10y}{25}$

12. If $Y-a=X$ and $\overline{X}+a=14$, then the value of $2\overline{Y}$ is

A. 7
B. 14
C. 28
D. 30

13. The daily two-sided travelling expense (in Rupees) of 6 students is 25, 50, 30, 43, 30, and 20. If the expense is the same for both sides, then the median of the daily two-sided travelling expense of 6 students is

A. Rs 30
B. Rs 33
C. Rs 36
D. Rs 43

14. The given table shows the age-wise distribution of students in a summer camp.

Age (Years)	Number of Students	Cumulative Frequency
5 – 9	20	20
10 – 14	70	90
15 – 19	100	190
20 – 24	45	235
25 – 29	15	250

The upper class boundary of the first quartile class is

A. 8.5
B. 9.0
C. 9.5
D. 10.5

AKU-EB Annual Examinations 2025 For Teaching & Learning only

15. Consider the given stem and leave diagram of a certain data.

1	5 6 7 8 8 9
2	0 1 2 4
3	2 9
4	5 6

The total number of values in the data is

A. 4
B. 9
C. 12
D. 14

16. The variance in the values $\frac{3a}{2}, \frac{3a}{2}, \frac{3a}{2}, \frac{3a}{2}, \frac{3a}{2}$ and $\frac{3a}{2}$ is

A. 0
B. $\frac{3a}{2}$
C. $\frac{9a}{4}$
D. $\frac{9a^2}{4}$

17. The variance of the data $-1, -2, 0, 1$ and 2 is

A. 0
B. 1.41
C. 2
D. 2.50

18. If the variance in the values $1, 2, -2$ and 3 is 3.5, then the standard deviation of these values will be

A. 1.75
B. 1.87
C. 7.0
D. 12.25

AKU-EB Annual Examinations 2025 for Teaching & Learning only

19. The variance in the values $1, 2, -2$ and 3 is 3.5. If each of these values is reduced to half, then the variance will be

A. 0.875
B. 1.750
C. 3.500
D. 7.000

20. Consider the given table representing the price of four commodities.

Commodity	Price on March 30, 2022	Price on June 30, 2022
W	110	125
X	98	105
Y	60	80
Z	52	60

By taking the price on March 30, 2022 as a base, the index number would be

A. 123.333
B. 115.625
C. 86.486
D. 81.081

AKU-EB Annual Examinations 2025 for Teaching & Learning only

21. In terms of Laspeyres and Paasche's indices, the fisher price index is

A. standard deviation.
B. geometric mean.
C. arithmetic mean.
D. harmonic mean.

22. The factorial expression $\frac{3!n!}{2!(n+1)!}$ is equal to

A. $\frac{3}{n+1}$
B. $\frac{3n}{n+1}$
C. $\frac{3}{2(n+1)}$
D. $\frac{3n}{2(n+1)}$

23. In a walk for a cause rally, the registration number of the participants consists of two letters followed by two digits from 1 to 9.

If the repetition of letters and digits are NOT allowed, then the possible registration numbers are

A. 361
B. 722
C. 46,800
D. 54,756

24. How many different arrangements can be made from the letters of the word "GOLDEN" If all letters are used together?

A. 6
B. 36
C. 720
D. 46,656

25. At a college function, 8 friends are lined up for a photograph. The number of possible arrangements of these friends for the group photograph will be

A. 8
B. 64
C. 5,040
D. 40,320

26. There are 5 boys and 3 girls in a group. Three people are to be selected from the group. In how many different ways they can be selected if one particular boy should be there?

A. 7
B. 9
C. 12
D. 21

27. There are 10 boys and 18 girls in a class. If two students are selected at random, then the probability that one boy and one girl are selected will be

A. $\frac{10}{21}$
B. $\frac{5}{21}$
C. $\frac{5}{14}$
D. $\frac{3}{14}$

AKU-EDC Annual Examinations 2025 for Teaching & Learning only

28. A fair die is rolled once and two balanced coins are tossed. The probability of getting a head on the first coin, a tail on the second coin and the die shows three will be

(**Note**: The plural of die is dice.)

A. $\frac{2}{4} \times \frac{1}{6}$

B. $\frac{1}{4} \times \frac{1}{6}$

C. $\frac{2}{4} \times \frac{2}{6}$

D. $\frac{1}{4} \times \frac{3}{6}$

29. In the given table, the correlation between the variables X and Y is

X	6	9	10	15
Y	0	4	7	13

A. strong positive.
B. strong negative.
C. perfect positive.
D. perfect negative.

30. If there is a perfect inverse correlation between two variables, then the value of coefficient of correlation r will be

A. -1

B. -0.5

C. 0.5

D. 1

AKU-EB Annual Examinations 2025 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2025
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2025
for Teaching & Learning only